

Converting Between Standard Form and Slope-Intercept Form

Two forms of the same line. Standard form is $Ax + By = C$; slope-intercept form is $y = mx + b$, where m is the slope and b is the y -intercept. Converting means solving for y (or clearing fractions and collecting x and y on one side) — the line never changes, only the way it is written. Show every step, and write your final answer on the answer line.

This workshop covers: Standard form to slope-intercept form, slope-intercept form to standard form, slope from a table or two points, and converting inside a context.

1. Write $2x + y = 7$ in slope-intercept form. Then state the slope and the y -intercept.

Answer: _____

2. Write $y = 3x - 5$ in standard form $Ax + By = C$, using integer coefficients with $A > 0$.

Answer: _____

3. Write $3x + 4y = 12$ in slope-intercept form. Give the slope as a fraction in lowest terms.

Answer: _____

4. A pool is being drained at a steady rate. The table gives the water level (in centimetres) after x hours.

x (hours)	0	2	4	6
y (cm)	30	24	18	12

- (a) Read the water level, in centimetres, after 4 hours from the table.

Answer: _____

- (b) Use the table to find the rate of change and the starting level, then write the relationship in slope-intercept form.

Answer: _____

- (c) Rewrite that equation in standard form with integer coefficients and $A > 0$.

Answer: _____

5. Write $y = -\frac{2}{5}x + 3$ in standard form with integer coefficients and $A > 0$. (Clear the fraction first.)

Answer: _____

6. Write $-6x + 2y = 10$ in slope-intercept form. Watch the sign of the x term when you divide.

Answer: _____

7. A line passes through the points $(2, 9)$ and $(6, 17)$.

(a) Find the slope of the line.

Answer: _____

(b) Write the equation of the line in slope-intercept form.

Answer: _____

(c) Rewrite that equation in standard form $Ax + By = C$ with integer coefficients.

Answer: _____

8. Write $y = \frac{1}{2}x - 4$ in standard form with integer coefficients and $A > 0$.

Answer: _____

9. A drama club sells child tickets for \$4 and adult tickets for \$6. The club takes in \$120 in total, so the ticket numbers satisfy

$$4x + 6y = 120,$$

where x is the number of child tickets and y is the number of adult tickets sold. Write this equation in slope-intercept form, and state the y -intercept.

Answer: _____

10. Sam earns \$15 for every lawn he mows and \$25 for every driveway he clears. He wants to earn \$300 this month, so his jobs satisfy

$$15x + 25y = 300,$$

where x is the number of lawns and y is the number of driveways.

(a) Write the equation in slope-intercept form.

Answer: _____

(b) The slope is a trade-off rate. If Sam mows 5 more lawns, how many *fewer* driveways does he need to clear to still reach \$300?

Answer: _____